

2.3 Linear Equations

Introduction We continue our search for solutions of first-order DEs by next examining linear equations. Linear differential equations are an especially “friendly” family of differential equations in that, given a linear equation, whether first-order or a higher-order kin, there is always a good possibility that we can find some sort of solution of the equation that we can look at.

A Definition The form of a linear first-order DE was given in (7) of Section 1.1. This form, the case when $n = 1$ in (6) of that section, is reproduced here for convenience.

Definition 2.3.1 Linear Equation

A first-order differential equation of the form

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad (1)$$

is said to be a **linear equation** in the dependent variable y .

When $g(x) = 0$, the linear equation (1) is said to be **homogeneous**; otherwise, it is **nonhomogeneous**.

Standard Form By dividing both sides of (1) by the lead coefficient $a_1(x)$ we obtain a more useful form, the **standard form**, of a linear equation

$$\frac{dy}{dx} + P(x)y = f(x). \quad (2)$$

We seek a solution of (2) on an interval I for which both functions P and f are continuous.

In the discussion that follows, we illustrate a property and a procedure and end up with a formula representing the form that every solution of (2) must have. But more than the formula, the property and the procedure are important, because these two concepts carry over to linear equations of higher order.

The Property The differential equation (2) has the property that its solution is the **sum** of the two solutions, $y = y_c + y_p$, where y_c is a solution of the associated homogeneous equation

$$\frac{dy}{dx} + P(x)y = 0 \quad (3)$$

and y_p is a particular solution of the nonhomogeneous equation (2). To see this, observe

$$\frac{d}{dx} [y_c + y_p] + P(x)[y_c + y_p] = \underbrace{\left[\frac{dy_c}{dx} + P(x)y_c \right]}_0 + \underbrace{\left[\frac{dy_p}{dx} + P(x)y_p \right]}_{f(x)} = f(x).$$

The Homogeneous DE The homogeneous equation (3) is also separable. This fact enables us to find y_c by writing (3) as

$$\frac{dy}{y} + P(x) dx = 0$$

and integrating. Solving for y gives $y_c = ce^{-\int P(x)dx}$. For convenience let us write $y_c = cy_1(x)$, where $y_1 = e^{-\int P(x)dx}$. The fact that $dy_1/dx + P(x)y_1 = 0$ will be used next to determine y_p .

The Nonhomogeneous DE We can now find a particular solution of equation (2) by a procedure known as **variation of parameters**. The basic idea here is to find a function u so that

$y_p = u(x)y_1(x) = u(x)e^{-\int P(x)dx}$ is a solution of (2). In other words, our assumption for y_p is the same as $y_c = cy_1(x)$ except that c is replaced by the “variable parameter” u . Substituting $y_p = uy_1$ into (2) gives

$$\begin{array}{ccc} \text{Product Rule} & & \text{zero} \\ \downarrow & & \downarrow \\ u \frac{dy_1}{dx} + y_1 \frac{du}{dx} + P(x)uy_1 = f(x) & \text{or} & u \left[\frac{dy_1}{dx} + P(x)y_1 \right] + y_1 \frac{du}{dx} = f(x) \end{array}$$

so that $y_1 \frac{du}{dx} = f(x)$.

Separating variables and integrating then gives

$$du = \frac{f(x)}{y_1(x)} dx \quad \text{and} \quad u = \int \frac{f(x)}{y_1(x)} dx.$$

From the definition of $y_1(x)$, we see $1/y_1(x) = e^{\int P(x)dx}$. Therefore

$$y_p = uy_1 = \left(\int \frac{f(x)}{y_1(x)} dx \right) e^{-\int P(x)dx} = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx,$$

$$\text{and} \quad y = y_c + y_p = ce^{-\int P(x)dx} + e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx. \quad (4)$$

Hence if (2) has a solution, it must be of form (4). Conversely, it is a straightforward exercise in differentiation to verify that (4) constitutes a one-parameter family of solutions of equation (2).

You should not memorize the formula given in (4). There is an equivalent but easier way of solving (2). If (4) is multiplied by

$$e^{\int P(x)dx} \quad (5)$$

$$\text{and then} \quad e^{\int P(x)dx} y = c + \int e^{\int P(x)dx} f(x) dx \quad (6)$$

$$\text{is differentiated,} \quad \frac{d}{dx} [e^{\int P(x)dx} y] = e^{\int P(x)dx} f(x), \quad (7)$$

$$\text{we get} \quad e^{\int P(x)dx} \frac{dy}{dx} + P(x)e^{\int P(x)dx} y = e^{\int P(x)dx} f(x). \quad (8)$$

Dividing the last result by $e^{\int P(x)dx}$ gives (2).

Method of Solution The recommended method of solving (2) actually consists of (6)–(8) worked in reverse order. In other words, if (2) is multiplied by (5), we get (8). The left side of (8) is recognized as the derivative of the product of $e^{\int P(x)dx}$ and y . This gets us to (7). We then integrate both sides of (7) to get the solution (6). Because we can solve (2) by integration after multiplication by $e^{\int P(x)dx}$, we call this function an **integrating factor** for the differential equation. For convenience we summarize these results. We again emphasize that you should not memorize formula (4) but work through the following procedure each time.

Guidelines for Solving a Linear First-Order Equation

- (i) Put a linear equation of form (1) into standard form (2) and then determine $P(x)$ and the integrating factor $e^{\int P(x)dx}$.
- (ii) Multiply (2) by the integrating factor. The left side of the resulting equation is automatically the derivative of the integrating factor and y . Write

$$\frac{d}{dx} [e^{\int P(x)dx} y] = e^{\int P(x)dx} f(x)$$

and then integrate both sides of this equation.

EXAMPLE 1 Solving a Linear DE

Solve $\frac{dy}{dx} - 3y = 6$.

SOLUTION This linear equation can be solved by separation of variables. Alternatively, since the equation is already in the standard form (2), we see that the integrating factor is $e^{\int(-3)dx} = e^{-3x}$. We multiply the equation by this factor and recognize that

$$e^{-3x} \frac{dy}{dx} - 3e^{-3x}y = 6e^{-3x} \text{ is the same as } \frac{d}{dx}[e^{-3x}y] = 6e^{-3x}.$$

Integrating both sides of the last equation gives $e^{-3x}y = -2e^{-3x} + c$. Thus a solution of the differential equation is $y = -2 + ce^{3x}$, $-\infty < x < \infty$. $\equiv$

When a_1, a_0 , and g in (1) are constants, the differential equation is autonomous. In Example 1, you can verify from the form $dy/dx = 3(y + 2)$ that -2 is a critical point and that it is unstable and a repeller. Thus a solution curve with an initial point either above or below the graph of the equilibrium solution $y = -2$ pushes away from this horizontal line as x increases.

Constant of Integration Notice in the general discussion and in Example 1 we disregarded a constant of integration in the evaluation of the indefinite integral in the exponent of $e^{\int P(x)dx}$. If you think about the laws of exponents and the fact that the integrating factor multiplies both sides of the differential equation, you should be able to answer why writing $\int P(x)dx + c$ is unnecessary. See Problem 46 in Exercises 2.3.

General Solution Suppose again that the functions P and f in (2) are continuous on a common interval I . In the steps leading to (4) we showed that if (2) has a solution on I , then it must be of the form given in (4). Conversely, it is a straightforward exercise in differentiation to verify that any function of the form given in (4) is a solution of the differential equation (2) on I . In other words, (4) is a one-parameter family of solutions of equation (2), and every solution of (2) defined on I is a member of this family. Consequently, we are justified in calling (4) the **general solution** of the differential equation on the interval I . Now by writing (2) in the normal form $y' = F(x, y)$ we can identify $F(x, y) = -P(x)y + f(x)$ and $\partial F/\partial y = -P(x)$. From the continuity of P and f on the interval I , we see that F and $\partial F/\partial y$ are also continuous on I . With Theorem 1.2.1 as our justification, we conclude that there exists one and only one solution of the initial-value problem

$$\frac{dy}{dx} + P(x)y = f(x), \quad y(x_0) = y_0 \quad (9)$$

defined on some interval I_0 containing x_0 . But when x_0 is in I , finding a solution of (9) is just a matter of finding an appropriate value of c in (4); that is, for each x_0 in I there corresponds a distinct c . In other words, the interval I_0 of existence and uniqueness in Theorem 1.2.1 for the initial-value problem (9) is the entire interval I .

EXAMPLE 2 General Solution

Solve $x \frac{dy}{dx} - 4y = x^6 e^x$.

SOLUTION By dividing by x we get the standard form

$$\frac{dy}{dx} - \frac{4}{x}y = x^5 e^x. \quad (10)$$

From this form we identify $P(x) = -4/x$ and $f(x) = x^5 e^x$ and observe that P and f are continuous on the interval $(0, \infty)$. Hence the integrating factor is

we can use $\ln x$ instead of $\ln |x|$ since $x > 0$
 $\downarrow$
 $e^{-4 \int dx/x} = e^{-4 \ln x} = e^{\ln x^{-4}} = x^{-4}.$

Here we have used the basic identity $b^{\log_b N} = N$, $N > 0$. Now we multiply (10) by x^{-4} ,

$$x^{-4} \frac{dy}{dx} - 4x^{-5}y = xe^x, \text{ and obtain } \frac{d}{dx}[x^{-4}y] = xe^x.$$

It follows from integration by parts that the general solution defined on $(0, \infty)$ is $x^{-4}y = xe^x - e^x + c$ or $y = x^5e^x - x^4e^x + cx^4$. $\equiv$

Singular Points Except in the case when the lead coefficient is 1, the recasting of equation (1) into the standard form (2) requires division by $a_1(x)$. Values of x for which $a_1(x) = 0$ are called **singular points** of the equation. Singular points are potentially troublesome. Specifically in (2), if $P(x)$ (formed by dividing $a_0(x)$ by $a_1(x)$) is discontinuous at a point, the discontinuity may carry over to functions in the general solution of the differential equation.

EXAMPLE 3 General Solution

Find the general solution of $(x^2 - 9) \frac{dy}{dx} + xy = 0$.

SOLUTION We write the differential equation in standard form

$$\frac{dy}{dx} + \frac{x}{x^2 - 9}y = 0 \quad (11)$$

and identify $P(x) = x/(x^2 - 9)$. Although P is continuous on $(-\infty, -3)$, on $(-3, 3)$, and on $(3, \infty)$, we shall solve the equation on the first and third intervals. On these intervals the integrating factor is

$$e^{\int x dx/(x^2 - 9)} = e^{\frac{1}{2} \int 2x dx/(x^2 - 9)} = e^{\frac{1}{2} \ln |x^2 - 9|} = \sqrt{x^2 - 9}.$$

After multiplying the standard form (11) by this factor, we get

$$\frac{d}{dx}[\sqrt{x^2 - 9}y] = 0 \text{ and integrating gives } \sqrt{x^2 - 9}y = c.$$

Thus for either $x > 3$ or $x < -3$, the general solution of the equation is $y = c/\sqrt{x^2 - 9}$. $\equiv$

Notice in the preceding example that $x = 3$ and $x = -3$ are singular points of the equation and that every function in the general solution $y = c/\sqrt{x^2 - 9}$ is discontinuous at these points. On the other hand, $x = 0$ is a singular point of the differential equation in Example 2, but the general solution $y = x^5e^x - x^4e^x + cx^4$ is noteworthy in that every function in this one-parameter family is continuous at $x = 0$ and is defined on the interval $(-\infty, \infty)$ and not just on $(0, \infty)$ as stated in the solution. However, the family $y = x^5e^x - x^4e^x + cx^4$ defined on $(-\infty, \infty)$ cannot be considered the general solution of the DE, since the singular point $x = 0$ still causes a problem. See Problems 41 and 42 in Exercises 2.3. We will study singular points for linear differential equations in greater depth in Section 5.2.

EXAMPLE 4 An Initial-Value Problem

Solve the initial-value problem $\frac{dy}{dx} + y = x$, $y(0) = 4$.

SOLUTION The equation is in standard form, and $P(x) = 1$ and $f(x) = x$ are continuous on the interval $(-\infty, \infty)$. The integrating factor is $e^{\int dx} = e^x$, and so integrating

$$\frac{d}{dx}[e^xy] = xe^x$$

gives $e^x y = xe^x - e^x + c$. Solving this last equation for y yields the general solution $y = x - 1 + ce^{-x}$. But from the initial condition we know that $y = 4$ when $x = 0$. Substituting these values in the general solution implies $c = 5$. Hence the solution of the problem is

$$y = x - 1 + 5e^{-x}, \quad -\infty < x < \infty. \quad (12) \equiv$$

Recall that the general solution of every linear first-order differential equation is a sum of two special solutions: y_c , the general solution of the associated homogeneous equation (3), and y_p , a particular solution of the nonhomogeneous equation (2). In Example 4 we identify $y_c = ce^{-x}$ and $y_p = x - 1$. **FIGURE 2.3.1**, obtained with the aid of a graphing utility, shows (12) in blue along with other representative solutions in the family $y = x - 1 + ce^{-x}$. It is interesting to observe that as x gets large, the graphs of *all* members of the family are close to the graph of $y_p = x - 1$, which is shown in green in Figure 2.3.1. This is because the contribution of $y_c = ce^{-x}$ to the values of a solution becomes negligible for increasing values of x . We say that $y_c = ce^{-x}$ is a **transient term** since $y_c \rightarrow 0$ as $x \rightarrow \infty$. While this behavior is not a characteristic of all general solutions of linear equations (see Example 2), the notion of a **transient** is often important in applied problems.

Discontinuous Coefficients In applications the coefficients $P(x)$ and $f(x)$ in (2) may be piecewise continuous functions. Such an equation is sometimes referred to as a **piecewise linear equation**. In the next example $f(x)$ is piecewise continuous on the interval $[0, \infty)$ with a single discontinuity, namely, a (finite) jump discontinuity at $x = 1$. We solve the problem in two parts corresponding to the two intervals over which $f(x)$ is defined; each part consists of a linear equation solvable by the method of this section. It is then possible to piece together the two solutions at $x = 1$ so that $y(x)$ is continuous on $[0, \infty)$.

EXAMPLE 5 An Initial-Value Problem

Solve $\frac{dy}{dx} + y = f(x)$, $y(0) = 0$ where $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x > 1. \end{cases}$

SOLUTION The graph of the discontinuous function f is shown in **FIGURE 2.3.2**. We solve the DE for $y(x)$ first on the interval $[0, 1]$ and then on the interval $(1, \infty)$. For $0 \leq x \leq 1$ we have

$$\frac{dy}{dx} + y = 1 \quad \text{or, equivalently,} \quad \frac{d}{dx}[e^x y] = e^x.$$

Integrating this last equation and solving for y gives $y = 1 + c_1 e^{-x}$. Since $y(0) = 0$, we must have $c_1 = -1$, and therefore $y = 1 - e^{-x}$, $0 \leq x \leq 1$. Then for $x > 1$, the equation

$$\frac{dy}{dx} + y = 0$$

leads to $y = c_2 e^{-x}$. Hence we can write

$$y = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1 \\ c_2 e^{-x}, & x > 1. \end{cases}$$

By appealing to the definition of continuity at a point it is possible to determine c_2 so that the foregoing function is continuous at $x = 1$. The requirement that $\lim_{x \rightarrow 1^+} y(x) = y(1)$ implies that $c_2 e^{-1} = 1 - e^{-1}$ or $c_2 = e - 1$. As seen in **FIGURE 2.3.3**, the piecewise defined function

$$y = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1 \\ (e - 1)e^{-x}, & x > 1 \end{cases} \quad (13)$$

is continuous on the interval $[0, \infty)$. ≡

It is worthwhile to think about (13) and Figure 2.3.3 a little bit; you are urged to read and answer Problem 44 in Exercises 2.3.

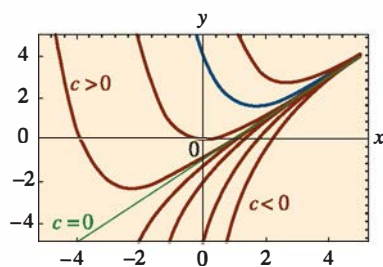


FIGURE 2.3.1 Some solutions of the DE in Example 4

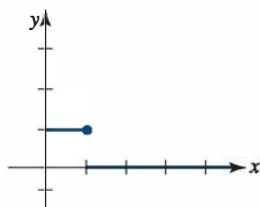


FIGURE 2.3.2 Discontinuous $f(x)$ in Example 5

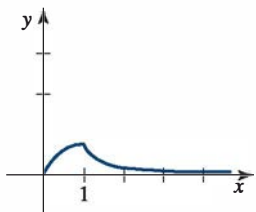


FIGURE 2.3.3 Graph of function in (13) of Example 5

Functions Defined by Integrals As pointed out in Section 2.2, some simple functions do not possess antiderivatives that are elementary functions, and integrals of these kinds of functions are called **nonelementary**. For example, you may have seen in calculus that $\int e^{x^2} dx$ and $\int \sin x^2 dx$ are nonelementary integrals. In applied mathematics some important functions are *defined* in terms of nonelementary integrals. Two such functions are the **error function** and **complementary error function**:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad \text{and} \quad \operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt. \quad (14)$$

Since $(2/\sqrt{\pi}) \int_0^\infty e^{-t^2} dt = 1$ it is seen from (14) that the error function $\operatorname{erf}(x)$ and the complementary error function $\operatorname{erfc}(x)$ are related by $\operatorname{erf}(x) + \operatorname{erfc}(x) = 1$. Because of its importance in areas such as probability and statistics, the error function has been extensively tabulated. Note that $\operatorname{erf}(0) = 0$ is one obvious functional value. Values of $\operatorname{erf}(x)$ can also be found using a CAS. Before working through the next example, you are urged to reread Example 5 and (i) of the *Remarks* in Section 2.2.

EXAMPLE 6 The Error Function

Solve the initial-value problem $\frac{dy}{dx} - 2xy = 2, y(0) = 1$.

SOLUTION Since the equation is already in standard form, we see that the integrating factor is e^{-x^2} , and so from

$$\frac{d}{dx} [e^{-x^2} y] = 2e^{-x^2} \quad \text{we get} \quad y = 2e^{x^2} \int_0^x e^{-t^2} dt + ce^{x^2}. \quad (15)$$

Applying $y(0) = 1$ to the last expression then gives $c = 1$. Hence, the solution to the problem is

$$y = 2e^{x^2} \int_0^x e^{-t^2} dt + e^{x^2} \quad \text{or} \quad y = e^{x^2} [1 + \sqrt{\pi} \operatorname{erf}(x)].$$

The graph of this solution, shown in blue in **FIGURE 2.3.4** among other members of the family defined by (15), was obtained with the aid of a computer algebra system (CAS).

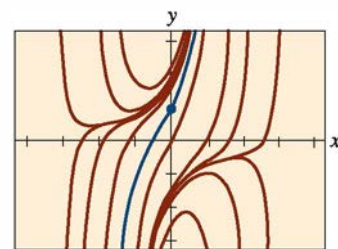


FIGURE 2.3.4 Some solutions of the DE in Example 6

Use of Computers Some computer algebra systems are capable of producing explicit solutions for some kinds of differential equations. For example, to solve the equation $y' + 2y = x$, we use the input commands

`DSolve[y'[x] + 2 y[x] == x, y[x], x]` (in *Mathematica*)

and `dsolve(diff(y(x), x) + 2*y(x) = x, y(x));` (in *Maple*)

Translated into standard symbols, the output of each program is $y = -\frac{1}{4} + \frac{1}{2}x + ce^{-2x}$.

Remarks

(i) Occasionally a first-order differential equation is not linear in one variable but is linear in the other variable. For example, the differential equation

$$\frac{dy}{dx} = \frac{1}{x + y^2}$$

is not linear in the variable y . But its reciprocal

$$\frac{dx}{dy} = x + y^2 \quad \text{or} \quad \frac{dx}{dy} - x = y^2$$